

Ishita Bhatt

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OBJECTIVE

Electrical and Computer Engineering student with a foundation in ML and robotics. Eager to work with ML to solve real world industry challenges.

EDUCATION

Shiv Nadar University, Delhi NCR

2024-2028

Pursuing B.Tech. in Electrical and Computer Engineering
(with Minor in Management)

CGPA: 8.2

- Relevant coursework:
 - ECE: Semiconductors, Embedded Systems, Digital Electronics, Analog Electronics, Python, C Programming, Calculus I & II, Probability and Statistics, Data Structures and Algorithms, Signals and Systems, Physics I & II
 - Management: Digitalization and Enterprise Risk Management, Social Entrepreneurship, Principles of Management

Sadhu Vaswani International School, Hyderabad

2022-2024

- Subjects: Physics, Chemistry, Mathematics, English, Entrepreneurship

EXPERIENCE

1. Research Intern

June 2026-present(10 week)

DRDO, Aerial Delivery Research & Development Establishment

Agra, Uttar Pradesh

- Working on Systems engineering, Data Analysis, feasibility study, data-driven analysis and design optimization.

PROJECTS (<https://github.com/IshitaBhatt486>)

1. Physics Engine (Python):

June 2025 – July 2025

- a. Designed and implemented a real-time 2D physics engine using Python and Pygame,
- b. Mathematical modelling of physical laws (conservation of momentum, vector math) to simulate realistic mass-based responses and damping to model energy losses and interactions at 60 FPS
- c. interactive UI allowing users to dynamically add objects, drag and throw them with the mouse, and observe real-time physics behaviour

2. Smart Greenhouse System using STM32:

Oct 2025 – Nov 2025

- a. Engineered a real-time control system using **STM32 (ARM Cortex-M) microcontroller** to **automate** environmental adjustments, **optimize** plant growth and ensure optimal resource utilization
- b. Integrated **analog and digital sensors** (DHT22, MQ135 and LDR), LCD screen, ADC and I2C protocol

3. Ball Bearing Sorting project:

Dec 2024 - Nov 2025

- a. Collaborated with mechanical engineering students to bridge the gap between software, electronics and mechanical actuation on a Ball Bearing Sorting project (for ASME eFX India Student Design Challenge)
- b. Worked on increasing speed and accuracy of the bot

4. Digital Memory Design & Verification

Oct 2025- Nov 2025

- a. Designed a synchronous 8-bit RAM and ROM module using Verilog HDL, implementing precise address decoding and read/write enable logic
- b. Developed comprehensive testbenches to verify data integrity across various memory addresses

SKILLS

- Skills: C, C++, Python, Verilog, SystemVerilog, MATLAB, Bash
- Frameworks and libraries: NumPy, Pandas, PyTorch, TensorFlow, Keras, OpenCV
- Other: RTOS, GitHub, Embedded Linux, Microcontrollers, LTSpice

OTHER

• Clubs:

1. Technical Lead, IEEE student chapter:

- **Captain of robotics team**, member of **Computer Society** and **Vehicular Technology society**, supervisor and advisor for various projects, organized and coordinated logistics and budgeting for **technical workshops and seminars**.
- Worked with microcontrollers (Arduino, ESP32, STM32, raspberry pi), computer vision and PCB design

2. ASME student chapter (American Society of Mechanical Engineers):

worked with 3D printing, prototyping, motors, sensors and robotics

- **Online Courses:** Semiconductor Fabrication 101(Intel), Linear Algebra(John Hopkins University)
- **Hobbies:** Horse Riding and Language learning